Ward Laboratories, Inc.

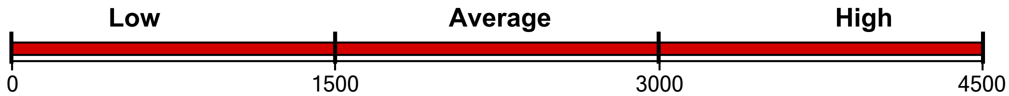
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003244

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T02-0-7.5**

PLFA Soil Microbial Community Analysis

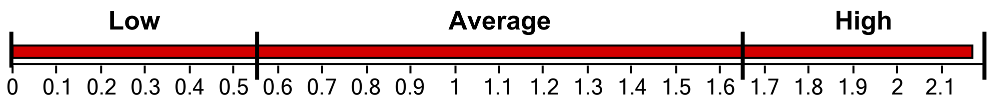
Functional Group Biomass

Total Living Microbial Biomass	5202.31 ng/g

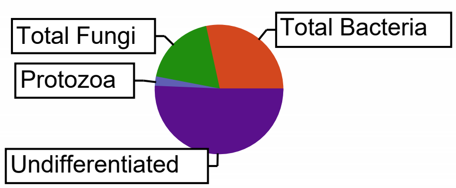


Functional Group Diversity

Functional Group Diversity Index	2.17



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1481.23	28.47
Gram (+)	541.06	10.40
Actinomycetes	183.73	3.53
Gram (-)	940.17	18.07
Rhizobia	0.00	0.00
Total Fungi	963.60	18.52
Arbuscular Mycorrhizal	158.07	3.04
Saprophytes	805.53	15.48
Protozoa	121.86	2.34
Undifferentiated	2635.63	50.66



Community Composition Ratios

Fungi:Bacteria	0.65
Predator:Prey	0.08
Gram(+):Gram(-)	0.58

Stress & Community Activity Ratios

Saturated:Unsaturated	1.13
Monounsaturated:Polyunsaturated	2.23
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	5.66

Reviewed by: Wootton, Stephanie

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

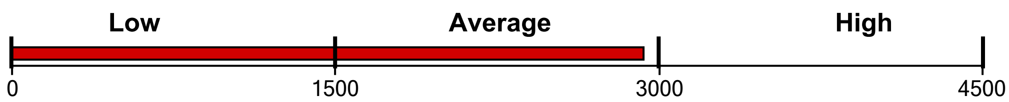
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003245

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T02-7.5-15**

PLFA Soil Microbial Community Analysis

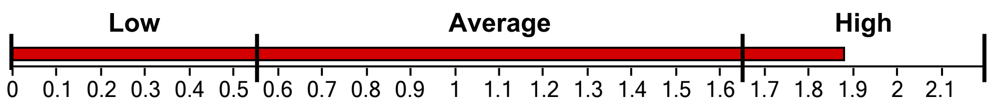
Functional Group Biomass

Total Living Microbial Biomass 2927.52 ng/g

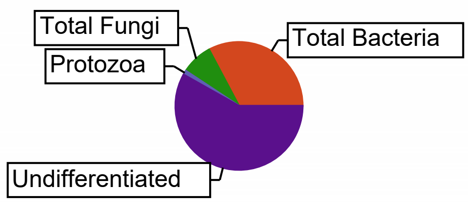


Functional Group Diversity

Functional Group Diversity Index 1.88



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	960.48	32.81
Gram (+)	302.94	10.35
Actinomycetes	123.55	4.22
Gram (-)	657.54	22.46
Rhizobia	0.00	0.00
Total Fungi	233.04	7.96
Arbuscular Mycorrhizal	54.77	1.87
Saprophytes	178.26	6.09
Protozoa	32.09	1.10
Undifferentiated	1701.92	58.14



Community Composition Ratios

Fungi:Bacteria	0.24
Predator:Prey	0.03
Gram(+):Gram(-)	0.46

Stress & Community Activity Ratios

Saturated:Unsaturated	1.82
Monounsaturated:Polyunsaturated	10.71
Pre 16:1 w7c:17:0 cyclo	1.07
Pre 18:1 w7c:19:0 cyclo	2.29

Reviewed by: Wootton, Stephanie

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

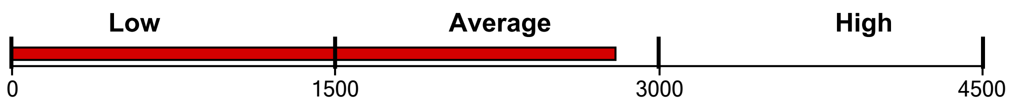
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003246

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T02-15-22.5**

PLFA Soil Microbial Community Analysis

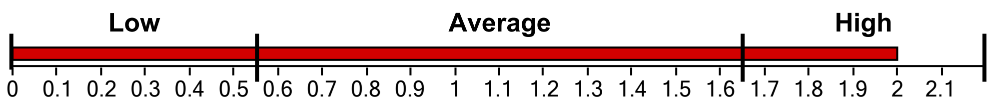
Functional Group Biomass

Total Living Microbial Biomass 2797.66 ng/g

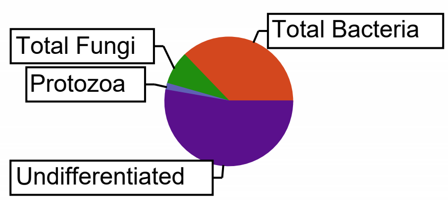


Functional Group Diversity

Functional Group Diversity Index 2.00



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1037.18	37.07
Gram (+)	372.69	13.32
Actinomycetes	159.53	5.70
Gram (-)	664.49	23.75
Rhizobia	0.00	0.00
Total Fungi	232.46	8.31
Arbuscular Mycorrhizal	42.54	1.52
Saprophytes	189.92	6.79
Protozoa	46.41	1.66
Undifferentiated	1469.15	52.51



Community Composition Ratios

Fungi:Bacteria	0.22
Predator:Prey	0.04
Gram(+):Gram(-)	0.56

Stress & Community Activity Ratios

Saturated:Unsaturated	1.76
Monounsaturated:Polyunsaturated	9.12
Pre 16:1 w7c:17:0 cyclo	1.54
Pre 18:1 w7c:19:0 cyclo	1.39

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

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Gram (+) : Gram (-)

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Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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WARD Laboratories, Inc.

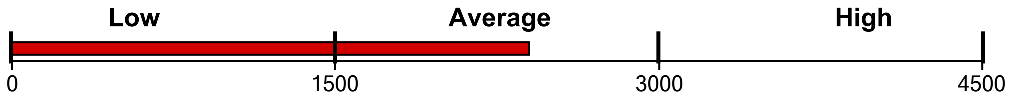
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003247

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T02-22.5-30**

PLFA Soil Microbial Community Analysis

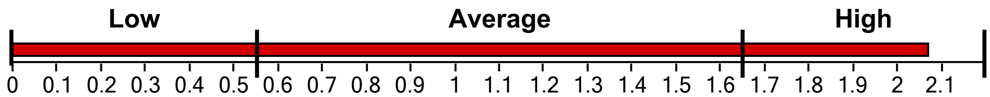
Functional Group Biomass

Total Living Microbial Biomass 2398.18 ng/g

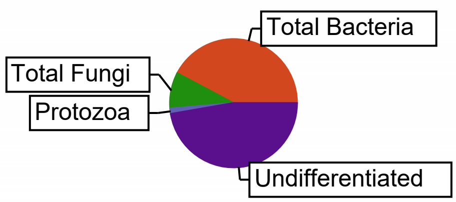


Functional Group Diversity

Functional Group Diversity Index 2.07



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1014.46	42.30
Gram (+)	356.65	14.87
Actinomycetes	168.26	7.02
Gram (-)	657.81	27.43
Rhizobia	0.00	0.00
Total Fungi	216.74	9.04
Arbuscular Mycorrhizal	45.29	1.89
Saprophytes	171.46	7.15
Protozoa	32.64	1.36
Undifferentiated	1134.33	47.30



Community Composition Ratios

Fungi:Bacteria	0.21
Predator:Prey	0.03
Gram(+):Gram(-)	0.54

Stress & Community Activity Ratios

Saturated:Unsaturated	1.92
Monounsaturated:Polyunsaturated	13.50
Pre 16:1 w7c:17:0 cyclo	1.48
Pre 18:1 w7c:19:0 cyclo	1.00

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

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Mono : Poly

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Laboratories, Inc.

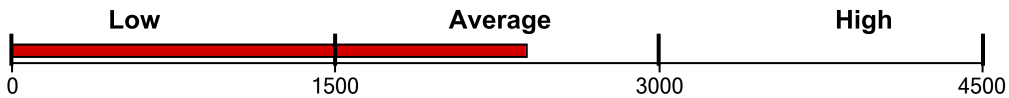
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003248

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T02-30-37.5**

PLFA Soil Microbial Community Analysis

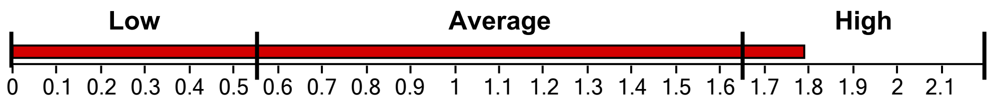
Functional Group Biomass

Total Living Microbial Biomass 2386.02 ng/g

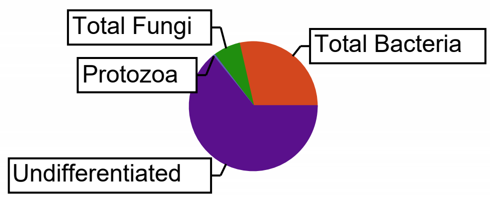


Functional Group Diversity

Functional Group Diversity Index 1.79



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	683.05	28.63
Gram (+)	277.49	11.63
Actinomycetes	142.06	5.95
Gram (-)	405.56	17.00
Rhizobia	0.00	0.00
Total Fungi	161.89	6.78
Arbuscular Mycorrhizal	27.03	1.13
Saprophytes	134.86	5.65
Protozoa	11.29	0.47
Undifferentiated	1529.79	64.11



Community Composition Ratios

Fungi:Bacteria	0.24
Predator:Prey	0.02
Gram(+):Gram(-)	0.68

Stress & Community Activity Ratios

Saturated:Unsaturated	2.18
Monounsaturated:Polyunsaturated	6.77
Pre 16:1 w7c:17:0 cyclo	1.36
Pre 18:1 w7c:19:0 cyclo	0.81

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

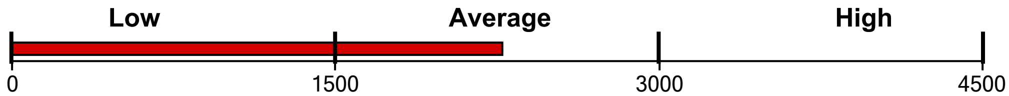
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003249

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-102-37-5-45**

PLFA Soil Microbial Community Analysis

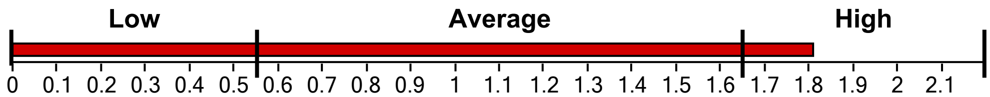
Functional Group Biomass

Total Living Microbial Biomass 2273.05 ng/g

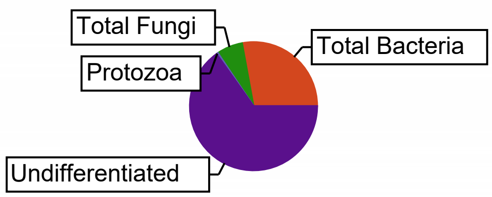


Functional Group Diversity

Functional Group Diversity Index 1.81



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	634.89	27.93
Gram (+)	308.25	13.56
Actinomycetes	188.39	8.29
Gram (-)	326.64	14.37
Rhizobia	0.00	0.00
Total Fungi	148.44	6.53
Arbuscular Mycorrhizal	26.18	1.15
Saprophytes	122.26	5.38
Protozoa	10.05	0.44
Undifferentiated	1479.67	65.10



Community Composition Ratios

Fungi:Bacteria	0.23
Predator:Prey	0.02
Gram(+):Gram(-)	0.94

Stress & Community Activity Ratios

Saturated:Unsaturated	2.92
Monounsaturated:Polyunsaturated	3.43
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.62

Reviewed by: Wootton, Stephanie

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

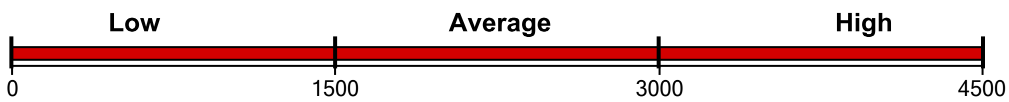
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003250

Results For: GLADE BOGAR
Field ID:
Sample ID: DO-T03-0-7.5

PLFA Soil Microbial Community Analysis

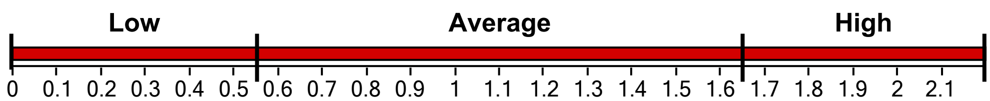
Functional Group Biomass

Total Living Microbial Biomass 6135.09 ng/g

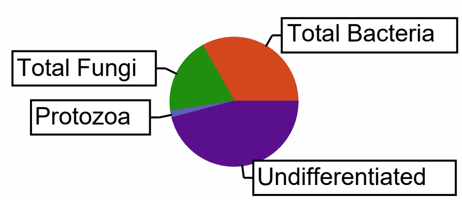


Functional Group Diversity

Functional Group Diversity Index 2.24



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	2040.66	33.26
Gram (+)	741.01	12.08
Actinomycetes	300.01	4.89
Gram (-)	1299.65	21.18
Rhizobia	0.00	0.00
Total Fungi	1177.17	19.19
Arbuscular Mycorrhizal	173.78	2.83
Saprophytes	1003.39	16.36
Protozoa	102.21	1.67
Undifferentiated	2815.04	45.88



Community Composition Ratios

Fungi:Bacteria	0.58
Predator:Prey	0.05
Gram(+):Gram(-)	0.57

Stress & Community Activity Ratios

Saturated:Unsaturated	1.03
Monounsaturated:Polyunsaturated	4.08
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	2.80

Reviewed by: Wootton, Stephanie

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Phone: 308-234-2418
www.wardlab.com
4007 Cherry Ave,
Kearney, Nebraska 68848

WARD Laboratories, Inc.

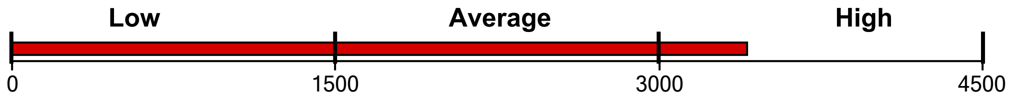
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003251

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T03-7.5-15**

PLFA Soil Microbial Community Analysis

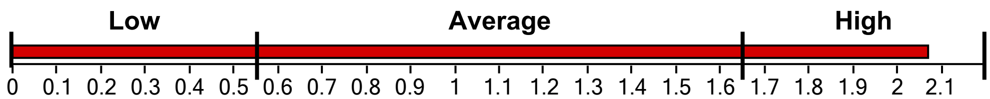
Functional Group Biomass

Total Living Microbial Biomass 3408.07 ng/g

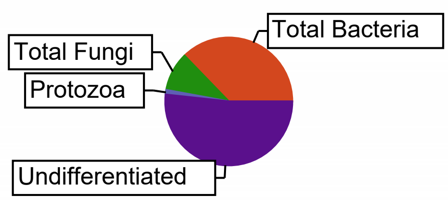


Functional Group Diversity

Functional Group Diversity Index 2.07



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1270.03	37.27
Gram (+)	536.08	15.73
Actinomycetes	252.57	7.41
Gram (-)	733.95	21.54
Rhizobia	0.00	0.00
Total Fungi	336.15	9.86
Arbuscular Mycorrhizal	68.09	2.00
Saprophytes	268.06	7.87
Protozoa	39.59	1.16
Undifferentiated	1762.30	51.71



Community Composition Ratios

Fungi:Bacteria	0.26
Predator:Prey	0.03
Gram(+):Gram(-)	0.73

Stress & Community Activity Ratios

Saturated:Unsaturated	1.97
Monounsaturated:Polyunsaturated	10.21
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.83

Reviewed by: **Wootton, Stephanie**

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Account ID: 98477

UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

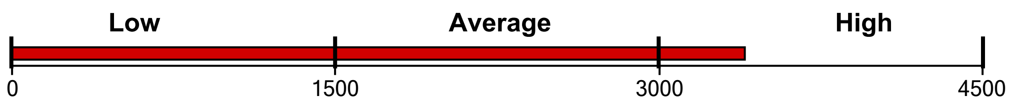
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003252

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T03-15-22.5**

PLFA Soil Microbial Community Analysis

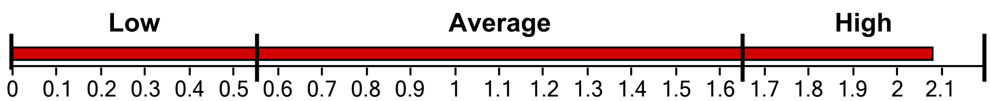
Functional Group Biomass

Total Living Microbial Biomass 3397.46 ng/g

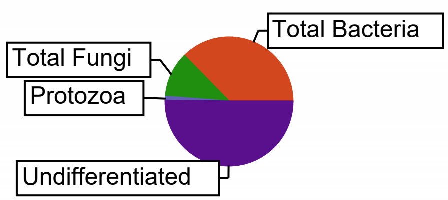


Functional Group Diversity

Functional Group Diversity Index 2.08



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1270.86	37.41
Gram (+)	450.36	13.26
Actinomycetes	208.86	6.15
Gram (-)	820.50	24.15
Rhizobia	0.00	0.00
Total Fungi	386.92	11.39
Arbuscular Mycorrhizal	56.94	1.68
Saprophytes	329.98	9.71
Protozoa	33.69	0.99
Undifferentiated	1705.98	50.21



Community Composition Ratios

Fungi:Bacteria	0.30
Predator:Prey	0.03
Gram(+):Gram(-)	0.55

Stress & Community Activity Ratios

Saturated:Unsaturated	1.65
Monounsaturated:Polyunsaturated	4.05
Pre 16:1 w7c:17:0 cyclo	1.47
Pre 18:1 w7c:19:0 cyclo	0.71

Reviewed by: Wootton, Stephanie

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Phone: 308-234-2418
www.wardlab.com
4007 Cherry Ave,
Kearney, Nebraska 68848

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

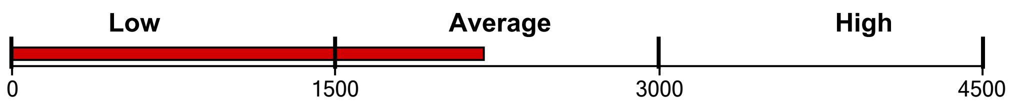
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003253

Results For: GLADE BOGAR
Field ID:
Sample ID: DO-T03-22.5-30

PLFA Soil Microbial Community Analysis

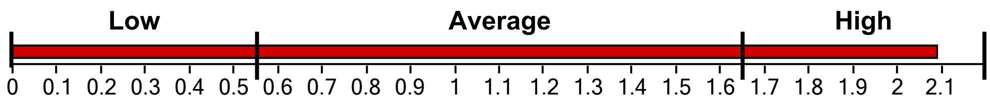
Functional Group Biomass

Total Living Microbial Biomass 2187.26 ng/g

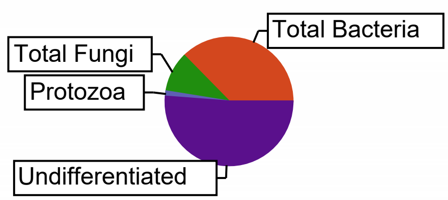


Functional Group Diversity

Functional Group Diversity Index 2.09



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	813.67	37.20
Gram (+)	335.14	15.32
Actinomycetes	183.15	8.37
Gram (-)	478.52	21.88
Rhizobia	0.00	0.00
Total Fungi	219.42	10.03
Arbuscular Mycorrhizal	40.49	1.85
Saprophytes	178.92	8.18
Protozoa	28.50	1.30
Undifferentiated	1114.67	50.96



Community Composition Ratios

Fungi:Bacteria	0.27
Predator:Prey	0.04
Gram(+):Gram(-)	0.70

Stress & Community Activity Ratios

Saturated:Unsaturated	2.44
Monounsaturated:Polyunsaturated	4.13
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.60

Reviewed by: Wootton, Stephanie

Ward Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

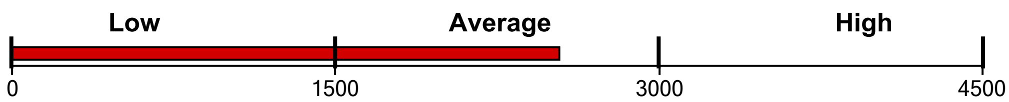
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003254

Results For: GLADE BOGAR
Field ID:
Sample ID: DO-T03-30-37.5

PLFA Soil Microbial Community Analysis

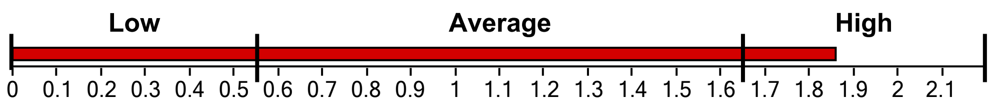
Functional Group Biomass

Total Living Microbial Biomass 2536.90 ng/g

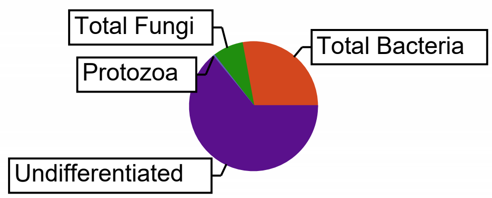


Functional Group Diversity

Functional Group Diversity Index 1.86



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	709.15	27.95
Gram (+)	359.10	14.16
Actinomycetes	207.61	8.18
Gram (-)	350.05	13.80
Rhizobia	0.00	0.00
Total Fungi	193.54	7.63
Arbuscular Mycorrhizal	27.68	1.09
Saprophytes	165.86	6.54
Protozoa	10.66	0.42
Undifferentiated	1623.55	64.00



Community Composition Ratios

Fungi:Bacteria	0.27
Predator:Prey	0.02
Gram(+):Gram(-)	1.03

Stress & Community Activity Ratios

Saturated:Unsaturated	2.66
Monounsaturated:Polyunsaturated	2.07
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.49

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

Saturated : Unsaturated

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Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003255

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T03-37-5-45**

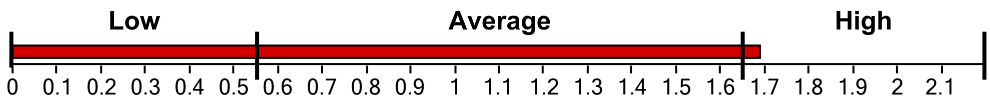
PLFA Soil Microbial Community Analysis

Functional Group Biomass

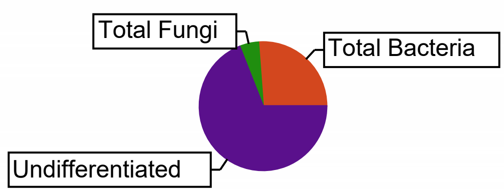
Total Living Microbial Biomass 1918.66 ng/g

Functional Group Diversity

Functional Group Diversity Index 1.69



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	502.85	26.21
Gram (+)	275.44	14.36
Actinomycetes	174.18	9.08
Gram (-)	227.41	11.85
Rhizobia	0.00	0.00
Total Fungi	92.13	4.80
Arbuscular Mycorrhizal	16.80	0.88
Saprophytes	75.33	3.93
Protozoa	0.00	0.00
Undifferentiated	1323.68	68.99



Community Composition Ratios

Fungi:Bacteria	0.18
Predator:Prey	0.00
Gram(+):Gram(-)	1.21

Stress & Community Activity Ratios

Saturated:Unsaturated	3.59
Monounsaturated:Polyunsaturated	2.92
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.41

Reviewed by: Wootton, Stephanie

WARD
Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

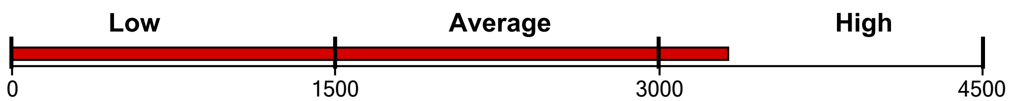
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003256

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-0-7.5**

PLFA Soil Microbial Community Analysis

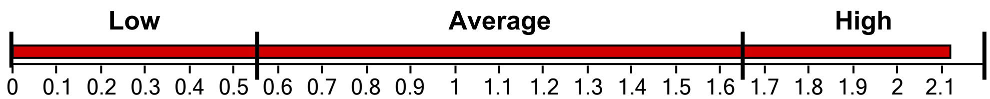
Functional Group Biomass

Total Living Microbial Biomass 3320.20 ng/g

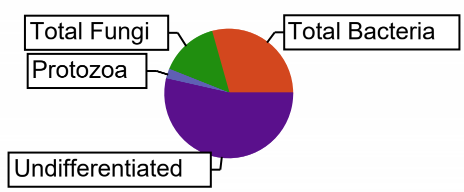


Functional Group Diversity

Functional Group Diversity Index 2.12



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	972.89	29.30
Gram (+)	366.38	11.03
Actinomycetes	155.66	4.69
Gram (-)	606.51	18.27
Rhizobia	0.00	0.00
Total Fungi	484.20	14.58
Arbuscular Mycorrhizal	77.68	2.34
Saprophytes	406.51	12.24
Protozoa	81.92	2.47
Undifferentiated	1771.47	53.35



Community Composition Ratios

Fungi:Bacteria	0.50
Predator:Prey	0.08
Gram(+):Gram(-)	0.60

Stress & Community Activity Ratios

Saturated:Unsaturated	1.26
Monounsaturated:Polyunsaturated	2.60
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	3.32

Reviewed by: **Wootton, Stephanie**

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

WARD Laboratories, Inc.

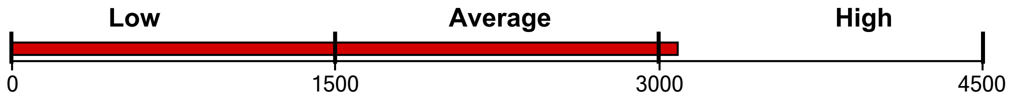
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003257

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-7.5-15**

PLFA Soil Microbial Community Analysis

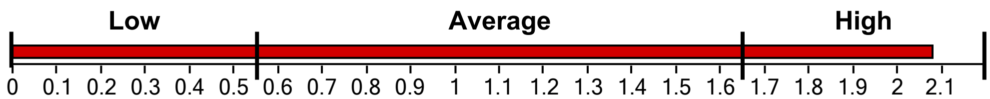
Functional Group Biomass

Total Living Microbial Biomass 3086.37 ng/g

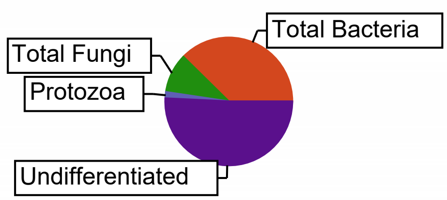


Functional Group Diversity

Functional Group Diversity Index 2.08



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1163.04	37.68
Gram (+)	428.66	13.89
Actinomycetes	213.99	6.93
Gram (-)	734.38	23.79
Rhizobia	0.00	0.00
Total Fungi	307.92	9.98
Arbuscular Mycorrhizal	68.92	2.23
Saprophytes	239.00	7.74
Protozoa	48.32	1.57
Undifferentiated	1567.11	50.77



Community Composition Ratios

Fungi:Bacteria	0.26
Predator:Prey	0.04
Gram(+):Gram(-)	0.58

Stress & Community Activity Ratios

Saturated:Unsaturated	1.48
Monounsaturated:Polyunsaturated	10.67
Pre 16:1 w7c:17:0 cyclo	2.18
Pre 18:1 w7c:19:0 cyclo	1.40

Reviewed by: Wootton, Stephanie

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Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

Saturated : Unsaturated

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Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Phone: 308-234-2418 www.wardlab.com 4007 Cherry Ave, Kearney, Nebraska 68848

Laboratories, Inc.

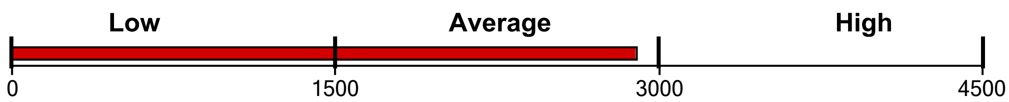
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003258

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-15-22.5**

PLFA Soil Microbial Community Analysis

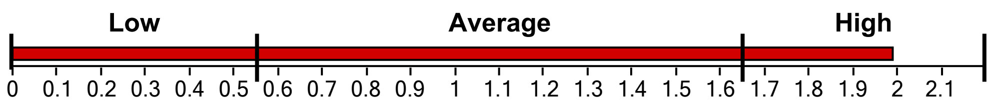
Functional Group Biomass

Total Living Microbial Biomass 2896.32 ng/g

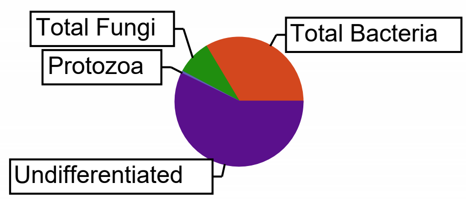


Functional Group Diversity

Functional Group Diversity Index 1.99



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	974.81	33.66
Gram (+)	456.19	15.75
Actinomycetes	252.73	8.73
Gram (-)	518.62	17.91
Rhizobia	0.00	0.00
Total Fungi	250.68	8.66
Arbuscular Mycorrhizal	49.89	1.72
Saprophytes	200.79	6.93
Protozoa	15.47	0.53
Undifferentiated	1655.36	57.15



Community Composition Ratios

Fungi:Bacteria	0.26
Predator:Prey	0.02
Gram(+):Gram(-)	0.88

Stress & Community Activity Ratios

Saturated:Unsaturated	2.40
Monounsaturated:Polyunsaturated	3.92
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.64

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
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0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

WARD Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

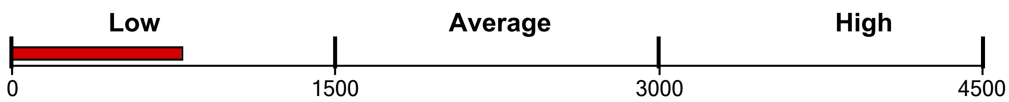
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003259

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-22.5-30**

PLFA Soil Microbial Community Analysis

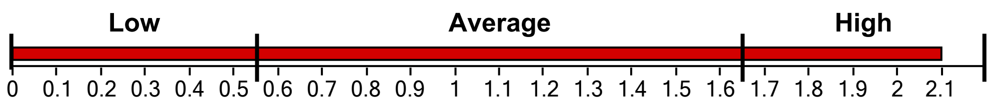
Functional Group Biomass

Total Living Microbial Biomass 788.51 ng/g

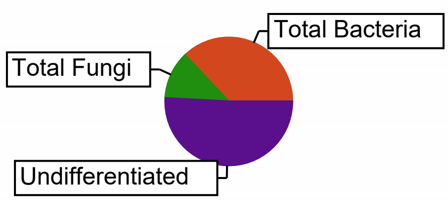


Functional Group Diversity

Functional Group Diversity Index 2.10



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	292.29	37.07
Gram (+)	130.81	16.59
Actinomycetes	67.19	8.52
Gram (-)	161.48	20.48
Rhizobia	0.00	0.00
Total Fungi	95.47	12.11
Arbuscular Mycorrhizal	15.43	1.96
Saprophytes	80.04	10.15
Protozoa	0.00	0.00
Undifferentiated	400.74	50.82



Community Composition Ratios

Fungi:Bacteria	0.33
Predator:Prey	0.00
Gram(+):Gram(-)	0.81

Stress & Community Activity Ratios

Saturated:Unsaturated	2.58
Monounsaturated:Polyunsaturated	4.31
Pre 16:1 w7c:17:0 cyclo	1.29
Pre 18:1 w7c:19:0 cyclo	0.50

Reviewed by: **Wootton, Stephanie**

Laboratories, Inc.

Diversity Index Ratings		
Total Biomass	**Diversity**	**Rating**
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings	
Scale	**Rating**
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings	
Scale	**Rating**
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings	
Scale	**Rating**
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003260

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-30-37.5**

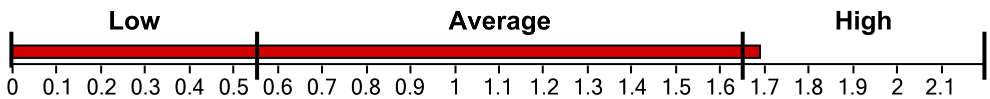
PLFA Soil Microbial Community Analysis

Functional Group Biomass

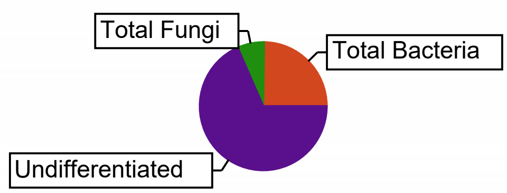
Total Living Microbial Biomass 1782.17 ng/g

Functional Group Diversity

Functional Group Diversity Index 1.69



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	442.13	24.81
Gram (+)	191.38	10.74
Actinomycetes	99.08	5.56
Gram (-)	250.74	14.07
Rhizobia	0.00	0.00
Total Fungi	121.26	6.80
Arbuscular Mycorrhizal	14.07	0.79
Saprophytes	107.19	6.01
Protozoa	0.00	0.00
Undifferentiated	1218.78	68.39



Community Composition Ratios

Fungi:Bacteria	0.27
Predator:Prey	0.00
Gram(+):Gram(-)	0.76

Stress & Community Activity Ratios

Saturated:Unsaturated	2.54
Monounsaturated:Polyunsaturated	2.44
Pre 16:1 w7c:17:0 cyclo	0.96
Pre 18:1 w7c:19:0 cyclo	0.27

Reviewed by: Wootton, Stephanie

Phone: 308-234-2418
www.wardlab.com
4007 Cherry Ave,
Kearney, Nebraska 68848

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Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

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Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

WARD Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

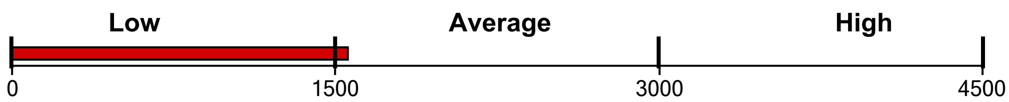
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003261

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T04-37-5-45**

PLFA Soil Microbial Community Analysis

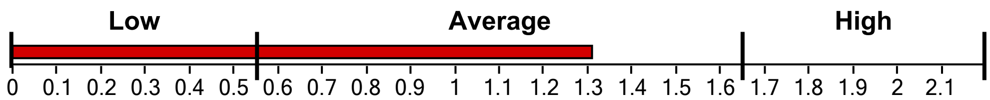
Functional Group Biomass

Total Living Microbial Biomass 1556.76 ng/g

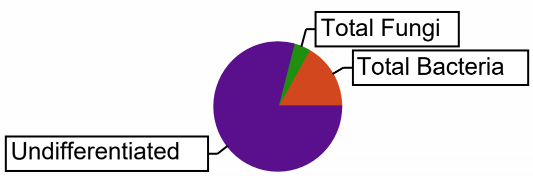


Functional Group Diversity

Functional Group Diversity Index 1.31



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	262.71	16.88
Gram (+)	113.94	7.32
Actinomycetes	55.59	3.57
Gram (-)	148.77	9.56
Rhizobia	0.00	0.00
Total Fungi	64.21	4.12
Arbuscular Mycorrhizal	0.00	0.00
Saprophytes	64.21	4.12
Protozoa	0.00	0.00
Undifferentiated	1229.84	79.00



Community Composition Ratios

Fungi:Bacteria	0.24
Predator:Prey	0.00
Gram(+):Gram(-)	0.77

Stress & Community Activity Ratios

Saturated:Unsaturated	3.25
Monounsaturated:Polyunsaturated	1.89
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.16

Reviewed by: **Wootton, Stephanie**

Laboratories, Inc.

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

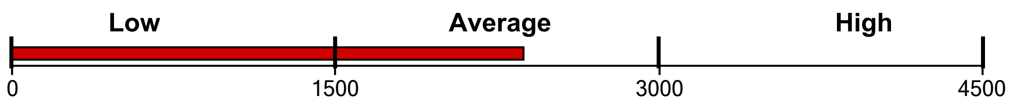
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003262

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-0-7.5**

PLFA Soil Microbial Community Analysis

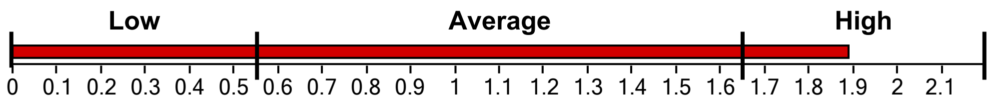
Functional Group Biomass

Total Living Microbial Biomass 2371.19 ng/g

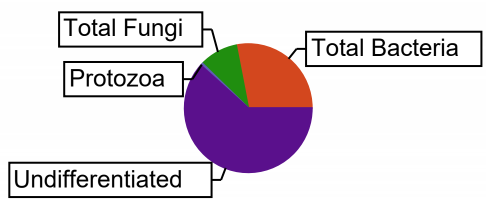


Functional Group Diversity

Functional Group Diversity Index 1.89



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	666.08	28.09
Gram (+)	256.50	10.82
Actinomycetes	131.26	5.54
Gram (-)	409.58	17.27
Rhizobia	0.00	0.00
Total Fungi	229.92	9.70
Arbuscular Mycorrhizal	32.15	1.36
Saprophytes	197.77	8.34
Protozoa	14.32	0.60
Undifferentiated	1460.87	61.61



Community Composition Ratios

Fungi:Bacteria	0.35
Predator:Prey	0.02
Gram(+):Gram(-)	0.63

Stress & Community Activity Ratios

Saturated:Unsaturated	1.70
Monounsaturated:Polyunsaturated	4.53
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	1.57

Reviewed by: **Wootton, Stephanie**

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3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
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Scale	Rating
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Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
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Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

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Stress and Community Activity Ratios

Saturated : Unsaturated

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Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

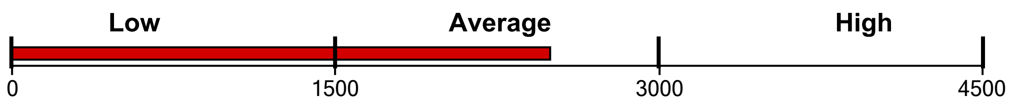
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003263

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-7.5-15**

PLFA Soil Microbial Community Analysis

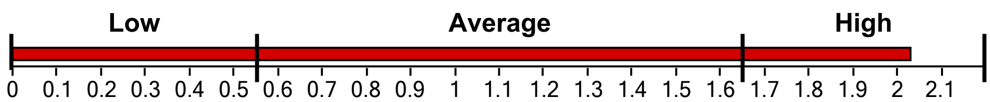
Functional Group Biomass

Total Living Microbial Biomass 2494.43 ng/g

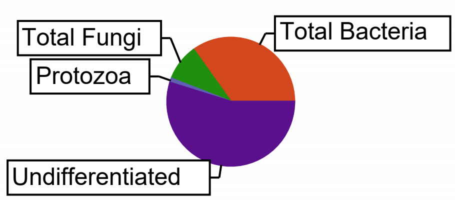


Functional Group Diversity

Functional Group Diversity Index 2.03



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	871.01	34.92
Gram (+)	360.60	14.46
Actinomycetes	188.74	7.57
Gram (-)	510.41	20.46
Rhizobia	0.00	0.00
Total Fungi	229.34	9.19
Arbuscular Mycorrhizal	47.96	1.92
Saprophytes	181.38	7.27
Protozoa	27.45	1.10
Undifferentiated	1366.63	54.79



Community Composition Ratios

Fungi:Bacteria	0.26
Predator:Prey	0.03
Gram(+):Gram(-)	0.71

Stress & Community Activity Ratios

Saturated:Unsaturated	1.97
Monounsaturated:Polyunsaturated	7.41
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.63

Reviewed by: Wootton, Stephanie

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

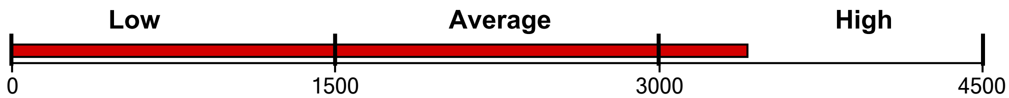
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003264

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-15-22.5**

PLFA Soil Microbial Community Analysis

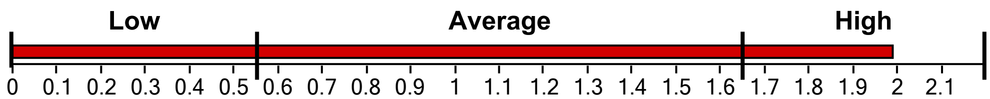
Functional Group Biomass

Total Living Microbial Biomass	3409.30 ng/g

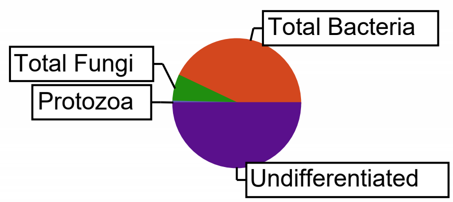


Functional Group Diversity

Functional Group Diversity Index	1.99



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1465.35	42.98
Gram (+)	626.75	18.38
Actinomycetes	342.19	10.04
Gram (-)	838.60	24.60
Rhizobia	0.00	0.00
Total Fungi	226.52	6.64
Arbuscular Mycorrhizal	46.93	1.38
Saprophytes	179.60	5.27
Protozoa	13.24	0.39
Undifferentiated	1704.18	49.99



Community Composition Ratios

Fungi:Bacteria	0.15
Predator:Prey	0.01
Gram(+):Gram(-)	0.75

Stress & Community Activity Ratios

Saturated:Unsaturated	2.85
Monounsaturated:Polyunsaturated	10.72
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.37

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

Account ID:	**98477**
	UC DAVIS BOGAR LAB - GLADE BOGAR
	1 SHIELDS AVENUE
	DAVIS, CA 95616

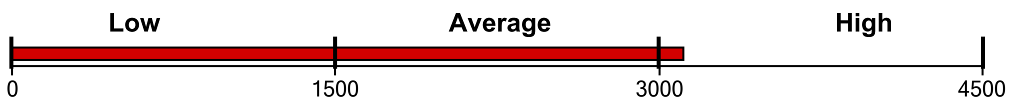
Report Type:	Biological
Invoice No.:	
Date Received:	8/20/2024
Date Reported:	9/27/2024
Lab ID:	B24-003265

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-22.5-30**

PLFA Soil Microbial Community Analysis

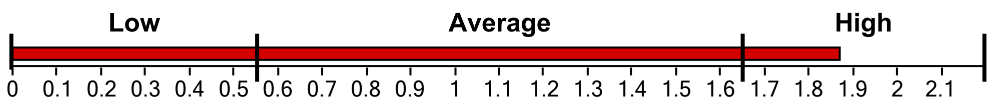
Functional Group Biomass

Total Living Microbial Biomass 3112.49 ng/g

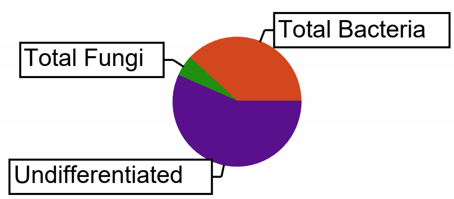


Functional Group Diversity

Functional Group Diversity Index 1.87



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1189.63	38.22
Gram (+)	636.56	20.45
Actinomycetes	304.96	9.80
Gram (-)	553.07	17.77
Rhizobia	0.00	0.00
Total Fungi	162.67	5.23
Arbuscular Mycorrhizal	34.81	1.12
Saprophytes	127.86	4.11
Protozoa	0.00	0.00
Undifferentiated	1760.19	56.55



Community Composition Ratios

Fungi:Bacteria	0.14
Predator:Prey	0.00
Gram(+):Gram(-)	1.15

Stress & Community Activity Ratios

Saturated:Unsaturated	3.91
Monounsaturated:Polyunsaturated	10.81
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.30

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

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Mono : Poly

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

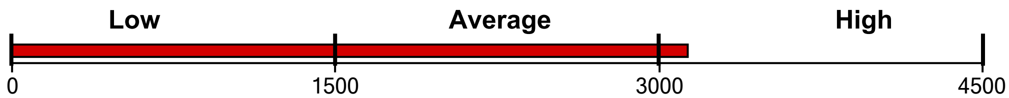
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003266

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-30-37.5**

PLFA Soil Microbial Community Analysis

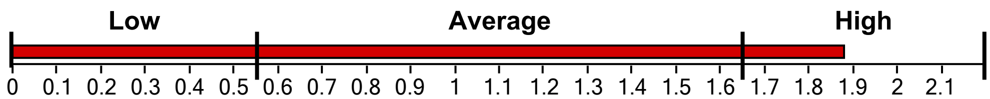
Functional Group Biomass

Total Living Microbial Biomass 3132.40 ng/g

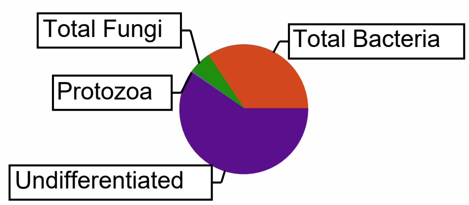


Functional Group Diversity

Functional Group Diversity Index 1.88



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1080.00	34.48
Gram (+)	608.81	19.44
Actinomycetes	316.60	10.11
Gram (-)	471.19	15.04
Rhizobia	0.00	0.00
Total Fungi	178.78	5.71
Arbuscular Mycorrhizal	34.50	1.10
Saprophytes	144.29	4.61
Protozoa	9.48	0.30
Undifferentiated	1864.13	59.51



Community Composition Ratios

Fungi:Bacteria	0.17
Predator:Prey	0.01
Gram(+):Gram(-)	1.29

Stress & Community Activity Ratios

Saturated:Unsaturated	3.61
Monounsaturated:Polyunsaturated	3.83
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.36

Reviewed by: **Wootton, Stephanie**

WARD Laboratories, Inc.

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Community Composition Ratios

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Mono : Poly

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

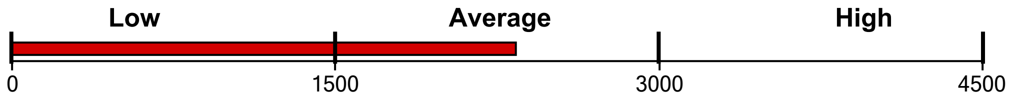
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003267

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T05-37-5-45**

PLFA Soil Microbial Community Analysis

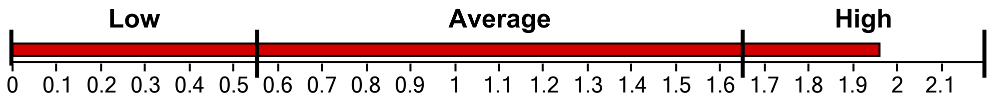
Functional Group Biomass

Total Living Microbial Biomass 2335.39 ng/g

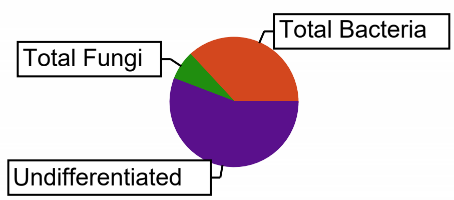


Functional Group Diversity

Functional Group Diversity Index 1.96



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	863.92	36.99
Gram (+)	516.25	22.11
Actinomycetes	278.13	11.91
Gram (-)	347.67	14.89
Rhizobia	0.00	0.00
Total Fungi	169.47	7.26
Arbuscular Mycorrhizal	26.29	1.13
Saprophytes	143.18	6.13
Protozoa	0.00	0.00
Undifferentiated	1302.00	55.75



Community Composition Ratios

Fungi:Bacteria	0.20
Predator:Prey	0.00
Gram(+):Gram(-)	1.48

Stress & Community Activity Ratios

Saturated:Unsaturated	3.99
Monounsaturated:Polyunsaturated	3.95
Pre 16:1 w7c:17:0 cyclo	None
Pre 18:1 w7c:19:0 cyclo	0.34

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Ward Laboratories, Inc.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

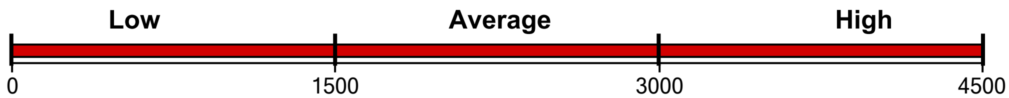
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003268

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T06-0-7.5**

PLFA Soil Microbial Community Analysis

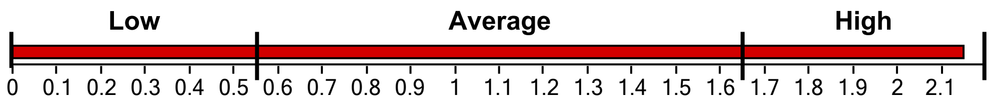
Functional Group Biomass

Total Living Microbial Biomass 5158.41 ng/g

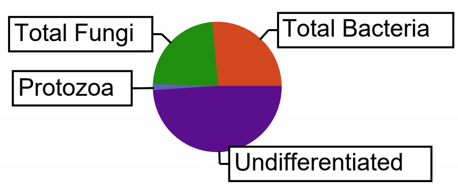


Functional Group Diversity

Functional Group Diversity Index 2.15



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1361.89	26.40
Gram (+)	554.61	10.75
Actinomycetes	153.13	2.97
Gram (-)	807.28	15.65
Rhizobia	0.00	0.00
Total Fungi	1187.70	23.02
Arbuscular Mycorrhizal	101.67	1.97
Saprophytes	1086.03	21.05
Protozoa	77.47	1.50
Undifferentiated	2521.29	48.88



Community Composition Ratios

Fungi:Bacteria	0.87
Predator:Prey	0.06
Gram(+):Gram(-)	0.69

Stress & Community Activity Ratios

Saturated:Unsaturated	1.04
Monounsaturated:Polyunsaturated	0.66
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	4.83

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

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Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Phone: 308-234-2418
www.wardlab.com
4007 Cherry Ave,
Kearney, Nebraska 68848

WARD Laboratories, Inc.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

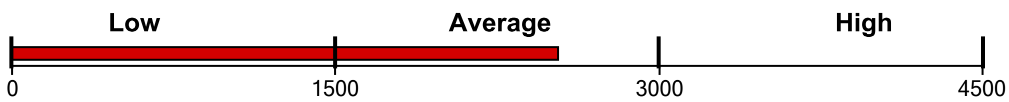
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003269

Results For: GLADE BOGAR
Field ID:
Sample ID: DO-T06-7.5-15

PLFA Soil Microbial Community Analysis

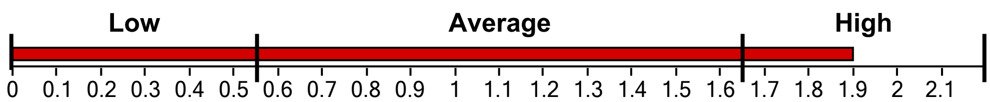
Functional Group Biomass

Total Living Microbial Biomass 2531.40 ng/g

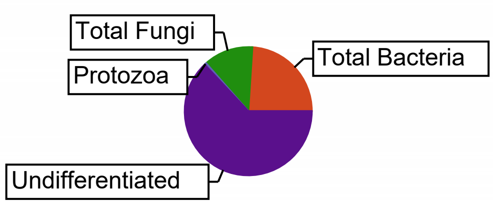


Functional Group Diversity

Functional Group Diversity Index 1.90



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	603.88	23.86
Gram (+)	289.87	11.45
Actinomycetes	132.72	5.24
Gram (-)	314.01	12.40
Rhizobia	0.00	0.00
Total Fungi	315.92	12.48
Arbuscular Mycorrhizal	28.48	1.13
Saprophytes	287.44	11.35
Protozoa	12.98	0.51
Undifferentiated	1584.80	62.61



Community Composition Ratios

Fungi:Bacteria	0.52
Predator:Prey	0.02
Gram(+):Gram(-)	0.92

Stress & Community Activity Ratios

Saturated:Unsaturated	2.00
Monounsaturated:Polyunsaturated	1.92
Pre 16:1 w7c:17:0 cyclo	1.80
Pre 18:1 w7c:19:0 cyclo	0.99

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

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Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Laboratories, Inc.

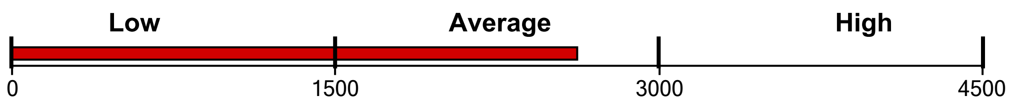
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003270

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T06-15-22.5**

PLFA Soil Microbial Community Analysis

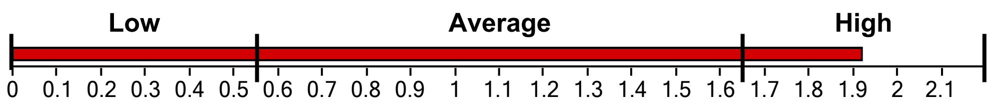
Functional Group Biomass

Total Living Microbial Biomass	2620.30 ng/g

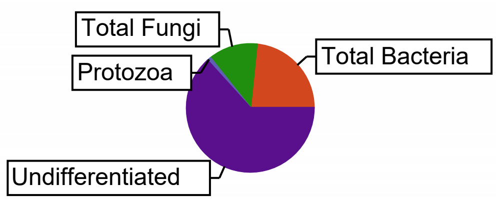


Functional Group Diversity

Functional Group Diversity Index	1.92



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	612.73	23.38
Gram (+)	355.50	13.57
Actinomycetes	159.88	6.10
Gram (-)	257.23	9.82
Rhizobia	0.00	0.00
Total Fungi	324.94	12.40
Arbuscular Mycorrhizal	31.07	1.19
Saprophytes	293.87	11.22
Protozoa	25.63	0.98
Undifferentiated	1657.01	63.24



Community Composition Ratios

Fungi:Bacteria	0.53
Predator:Prey	0.04
Gram(+):Gram(-)	1.38

Stress & Community Activity Ratios

Saturated:Unsaturated	2.34
Monounsaturated:Polyunsaturated	1.90
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.94

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

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Community Composition Ratios

Fungi : Bacteria

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Stress and Community Activity Ratios

Saturated : Unsaturated

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Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

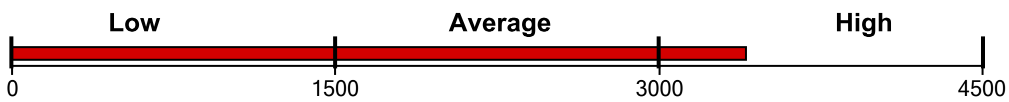
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003271

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T06-22.5-30**

PLFA Soil Microbial Community Analysis

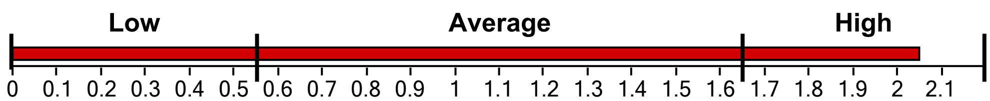
Functional Group Biomass

Total Living Microbial Biomass 3402.67 ng/g

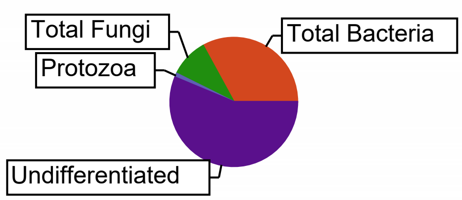


Functional Group Diversity

Functional Group Diversity Index 2.05



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1124.26	33.04
Gram (+)	629.99	18.51
Actinomycetes	320.67	9.42
Gram (-)	494.27	14.53
Rhizobia	0.00	0.00
Total Fungi	330.25	9.71
Arbuscular Mycorrhizal	54.19	1.59
Saprophytes	276.06	8.11
Protozoa	34.76	1.02
Undifferentiated	1913.40	56.23



Community Composition Ratios

Fungi:Bacteria	0.29
Predator:Prey	0.03
Gram(+):Gram(-)	1.27

Stress & Community Activity Ratios

Saturated:Unsaturated	2.37
Monounsaturated:Polyunsaturated	3.13
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.95

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

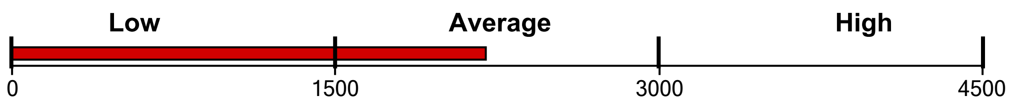
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003272

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T06-30-37.5**

PLFA Soil Microbial Community Analysis

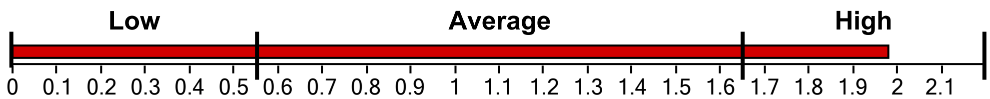
Functional Group Biomass

Total Living Microbial Biomass 2196.33 ng/g

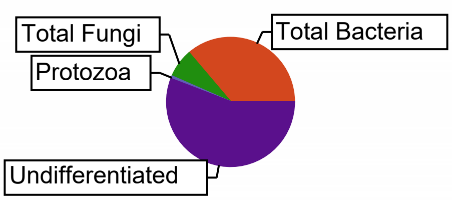


Functional Group Diversity

Functional Group Diversity Index 1.98



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	796.91	36.28
Gram (+)	417.43	19.01
Actinomycetes	226.69	10.32
Gram (-)	379.48	17.28
Rhizobia	0.00	0.00
Total Fungi	158.64	7.22
Arbuscular Mycorrhizal	27.89	1.27
Saprophytes	130.76	5.95
Protozoa	15.07	0.69
Undifferentiated	1225.70	55.81



Community Composition Ratios

Fungi:Bacteria	0.20
Predator:Prey	0.02
Gram(+):Gram(-)	1.10

Stress & Community Activity Ratios

Saturated:Unsaturated	2.43
Monounsaturated:Polyunsaturated	4.43
Pre 16:1 w7c:17:0 cyclo	1.46
Pre 18:1 w7c:19:0 cyclo	0.65

Reviewed by: **Wootton, Stephanie**

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

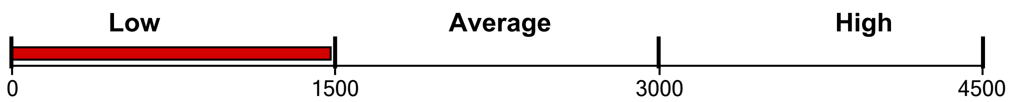
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003273

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T06-37-5-45**

PLFA Soil Microbial Community Analysis

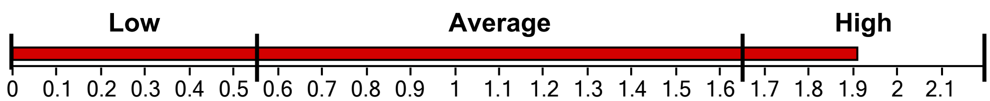
Functional Group Biomass

Total Living Microbial Biomass 1477.28 ng/g

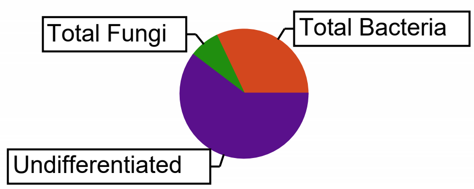


Functional Group Diversity

Functional Group Diversity Index 1.91



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	474.49	32.12
Gram (+)	262.07	17.74
Actinomycetes	140.64	9.52
Gram (-)	212.41	14.38
Rhizobia	0.00	0.00
Total Fungi	112.24	7.60
Arbuscular Mycorrhizal	19.06	1.29
Saprophytes	93.18	6.31
Protozoa	0.00	0.00
Undifferentiated	890.55	60.28



Community Composition Ratios

Fungi:Bacteria	0.24
Predator:Prey	0.00
Gram(+):Gram(-)	1.23

Stress & Community Activity Ratios

Saturated:Unsaturated	3.46
Monounsaturated:Polyunsaturated	4.13
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.49

Reviewed by: Wootton, Stephanie

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Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

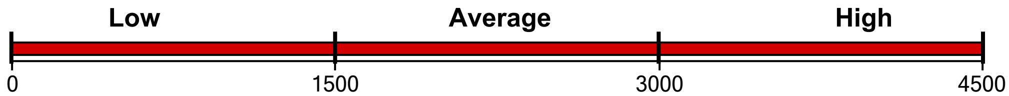
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003274

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-0-7.5**

PLFA Soil Microbial Community Analysis

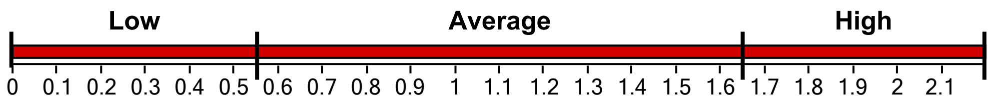
Functional Group Biomass

Total Living Microbial Biomass 5620.31 ng/g

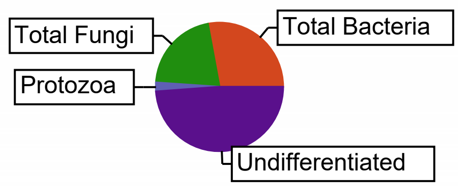


Functional Group Diversity

Functional Group Diversity Index 2.20



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1570.77	27.95
Gram (+)	651.35	11.59
Actinomycetes	204.56	3.64
Gram (-)	919.43	16.36
Rhizobia	0.00	0.00
Total Fungi	1177.81	20.96
Arbuscular Mycorrhizal	159.05	2.83
Saprophytes	1018.76	18.13
Protozoa	126.69	2.25
Undifferentiated	2745.04	48.84



Community Composition Ratios

Fungi:Bacteria	0.75
Predator:Prey	0.08
Gram(+):Gram(-)	0.71

Stress & Community Activity Ratios

Saturated:Unsaturated	1.09
Monounsaturated:Polyunsaturated	1.67
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	4.23

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

WARD Laboratories, Inc.

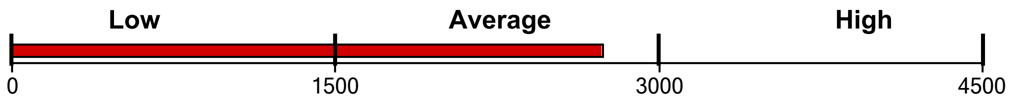
Account ID:	**98477**	**Report Type:**	Biological
	UC DAVIS BOGAR LAB - GLADE BOGAR	**Invoice No.:**	
	1 SHIELDS AVENUE	**Date Received:**	8/20/2024
	DAVIS, CA 95616	**Date Reported:**	9/27/2024
		Lab ID:	B24-003275

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-7.5-15**

PLFA Soil Microbial Community Analysis

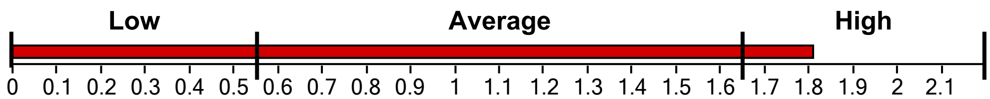
Functional Group Biomass

Total Living Microbial Biomass 2739.73 ng/g

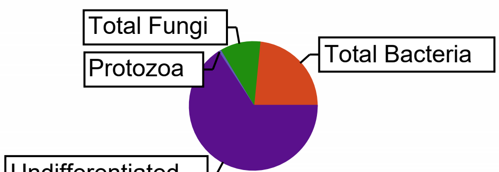


Functional Group Diversity

Functional Group Diversity Index 1.81



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	639.57	23.34
Gram (+)	326.68	11.92
Actinomycetes	128.15	4.68
Gram (-)	312.89	11.42
Rhizobia	0.00	0.00
Total Fungi	278.16	10.15
Arbuscular Mycorrhizal	36.84	1.34
Saprophytes	241.33	8.81
Protozoa	16.56	0.60
Undifferentiated	1794.44	65.50



Community Composition Ratios

Fungi:Bacteria	0.43
Predator:Prey	0.03
Gram(+):Gram(-)	1.04

Stress & Community Activity Ratios

Saturated:Unsaturated	2.51
Monounsaturated:Polyunsaturated	2.23
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	1.10

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

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Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: 98477

UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

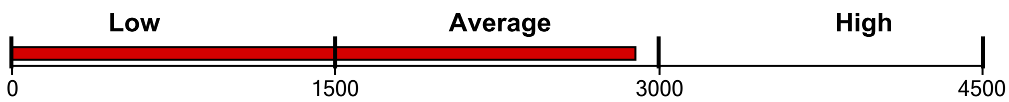
Report Type:	Biological
Invoice No.:	
Date Received:	8/20/2024
Date Reported:	9/27/2024
Lab ID:	B24-003276

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-15-22.5**

PLFA Soil Microbial Community Analysis

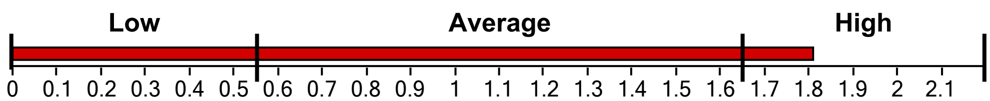
Functional Group Biomass

Total Living Microbial Biomass 2890.26 ng/g

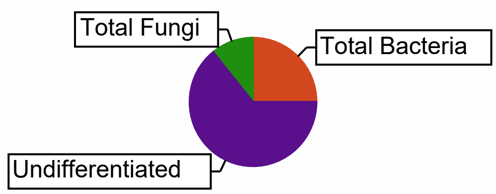


Functional Group Diversity

Functional Group Diversity Index 1.81



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	723.67	25.04
Gram (+)	343.34	11.88
Actinomycetes	117.18	4.05
Gram (-)	380.33	13.16
Rhizobia	0.00	0.00
Total Fungi	311.02	10.76
Arbuscular Mycorrhizal	32.55	1.13
Saprophytes	278.47	9.63
Protozoa	0.00	0.00
Undifferentiated	1855.57	64.20



Community Composition Ratios

Fungi:Bacteria	0.43
Predator:Prey	0.00
Gram(+):Gram(-)	0.90

Stress & Community Activity Ratios

Saturated:Unsaturated	2.34
Monounsaturated:Polyunsaturated	3.76
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.49

Reviewed by: Wootton, Stephanie

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Community Composition Ratios

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Stress and Community Activity Ratios

Saturated : Unsaturated

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Mono : Poly

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

WARD Laboratories, Inc.

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

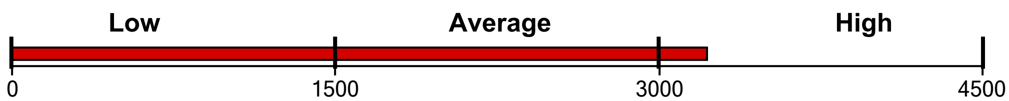
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003277

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-22.5-30**

PLFA Soil Microbial Community Analysis

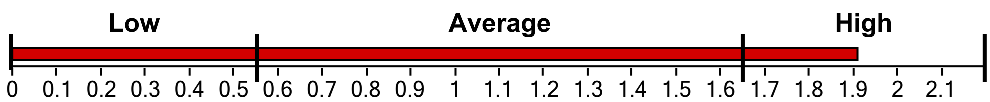
Functional Group Biomass

Total Living Microbial Biomass	3222.14 ng/g

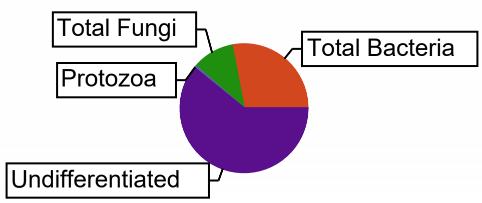


Functional Group Diversity

Functional Group Diversity Index	1.91



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	904.94	28.09
Gram (+)	393.59	12.22
Actinomycetes	155.75	4.83
Gram (-)	511.36	15.87
Rhizobia	0.00	0.00
Total Fungi	340.53	10.57
Arbuscular Mycorrhizal	49.76	1.54
Saprophytes	290.77	9.02
Protozoa	15.98	0.50
Undifferentiated	1960.69	60.85



Community Composition Ratios

Fungi:Bacteria	0.38
Predator:Prey	0.02
Gram(+):Gram(-)	0.77

Stress & Community Activity Ratios

Saturated:Unsaturated	2.45
Monounsaturated:Polyunsaturated	3.07
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.44

Reviewed by: Wootton, Stephanie

WARD Laboratories, Inc.

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

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Scale	Rating
< 0.05	Very Poor
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0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

Account ID: **98477**
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

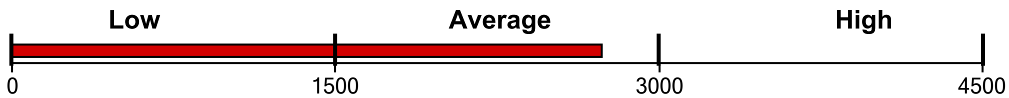
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003278

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-30-37.5**

PLFA Soil Microbial Community Analysis

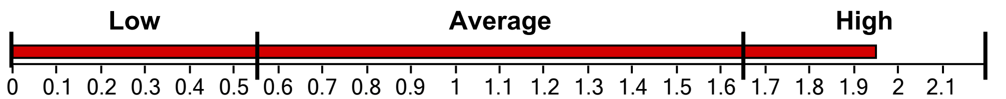
Functional Group Biomass

Total Living Microbial Biomass 2733.93 ng/g

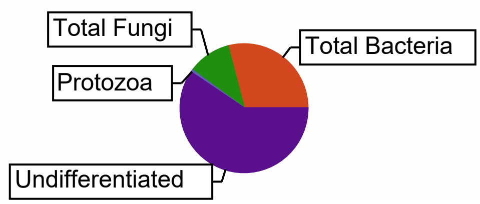


Functional Group Diversity

Functional Group Diversity Index 1.95



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	795.75	29.11
Gram (+)	365.35	13.36
Actinomycetes	146.83	5.37
Gram (-)	430.40	15.74
Rhizobia	0.00	0.00
Total Fungi	297.65	10.89
Arbuscular Mycorrhizal	43.87	1.60
Saprophytes	253.77	9.28
Protozoa	16.31	0.60
Undifferentiated	1624.22	59.41



Community Composition Ratios

Fungi:Bacteria	0.37
Predator:Prey	0.02
Gram(+):Gram(-)	0.85

Stress & Community Activity Ratios

Saturated:Unsaturated	2.29
Monounsaturated:Polyunsaturated	2.46
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.50

Reviewed by: **Wootton, Stephanie**

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Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

WARD Laboratories, Inc.

Account ID: 98477
UC DAVIS BOGAR LAB - GLADE BOGAR
1 SHIELDS AVENUE
DAVIS, CA 95616

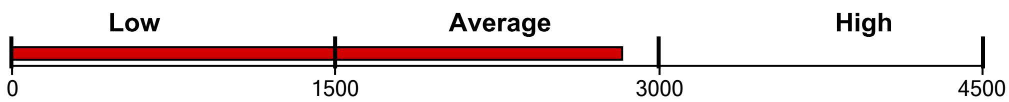
Report Type: Biological
Invoice No.:
Date Received: 8/20/2024
Date Reported: 9/27/2024
Lab ID: B24-003279

Results For: **GLADE BOGAR**
Field ID:
Sample ID: **DO-T07-37-5-45**

PLFA Soil Microbial Community Analysis

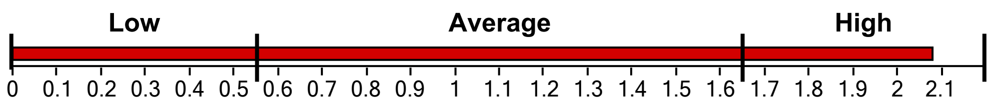
Functional Group Biomass

Total Living Microbial Biomass 2828.21 ng/g

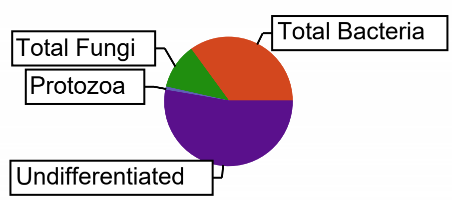


Functional Group Diversity

Functional Group Diversity Index 2.08



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	992.49	35.09
Gram (+)	476.39	16.84
Actinomycetes	197.89	7.00
Gram (-)	516.10	18.25
Rhizobia	0.00	0.00
Total Fungi	323.26	11.43
Arbuscular Mycorrhizal	52.82	1.87
Saprophytes	270.44	9.56
Protozoa	21.64	0.77
Undifferentiated	1490.82	52.71



Community Composition Ratios

Fungi:Bacteria	0.33
Predator:Prey	0.02
Gram(+):Gram(-)	0.92

Stress & Community Activity Ratios

Saturated:Unsaturated	2.54
Monounsaturated:Polyunsaturated	2.84
Pre 16:1 w7c:17:0 cyclo	0.00
Pre 18:1 w7c:19:0 cyclo	0.45

Reviewed by: Wootton, Stephanie

Laboratories, Inc.

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